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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2024

ME1104-1 – THERMAL ENGINEERING
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **An isolated system**
A) is a specified region where transfer of energy and /or mass take place
B) is a region of constant mass and only energy is allowed to cross the boundaries
C) cannot transfer either energy or mass to or from the surroundings
D) is one in which mass within the system is not necessarily constant
2. **Which of the following is an intensive property of a thermodynamic system?**
A) Volume
B) Temperature
C) Mass
D) Energy
3. **Absolute zero temperature is taken as**
A) -273°C
B) 273°C
C) 237°C
D) 373°C
4. **The characteristic equation of gases $pV = mRT$ holds good for**
A) monoatomic gases
B) diatomic gases
C) real gases
D) ideal gases
5. **In the polytropic process equation $pv^n = \text{constant}$, if $n=1$, the process is termed as**
A) Isochoric
B) Isobaric
C) Isothermal
D) Isentropic
6. **The adiabatic index (γ) is equal to the**
A) sum of two specific heats
B) difference of two specific heats
C) product of two specific heats
D) ratio of two specific heats
7. **Second law of thermodynamics defines**
A) heat
B) work
C) enthalpy
D) entropy
8. **Choose the correct statement**
A) Specific volume of water decreases on melting
B) Boiling point of water decreases with increasing pressure
C) Specific volume of CO_2 increases on freezing
D) Freezing temperature of water increases with increasing pressure
9. **Choose the correct statement**
A) The process of passing from liquid to vapor is condensation
B) An isothermal line is also a constant pressure line during wet region
C) Pressure and temperature are independent during phase change
D) The term dryness fraction is used to describe the fraction by mass of liquid in the mixture of liquid water and water vapour.
10. **With the increase in pressure**
A) boiling point of water increases, and enthalpy of evaporation increases
B) boiling point of water increases, and enthalpy of evaporation decreases
C) boiling point of water decreases, and enthalpy of evaporation increases
D) boiling point of water decreases, and enthalpy of evaporation decreases

11. Rankine cycle efficiency of a good steam power plant may be in the range of
 - A) 90 to 95%
 - B) 70 to 80%
 - C) 35 to 45%
 - D) 15 to 20%
12. Rankine efficiency of a steam power plant
 - A) Improves in summer as compared to that in winter
 - B) Improves in winter as compared to that in summer
 - C) Is unaffected by climate conditions
 - D) Can not be predicted
13. Regenerative heating i.e. bleeding steam to reheat feed water to boiler
 - A) decreases thermal efficiency of the cycle
 - B) does not affect thermal efficiency of the cycle
 - C) may decrease or increase thermal efficiency of the cycle depending upon the point of extraction of steam
 - D) increases thermal efficiency of the cycle
14. The air standard Otto cycle comprises of
 - A) two isobaric and two isochoric processes
 - B) two isothermal and two isobaric processes
 - C) two isobaric and two isentropic processes
 - D) two isochoric and two isentropic processes
15. In a Diesel cycle, compression ratio r , fuel cut-off ratio r_c and expansion ratio r_e are related as
 - A) $r = r_c \times r_e$
 - B) $r_c = r \times r_e$
 - C) $r_e = r \times r_c$
 - D) $r = r_c + r_e$
16. Thermal efficiency of Gas turbine power plant compared to Diesel power plant is
 - A) higher
 - B) lower
 - C) same
 - D) may be higher or lower
17. The function of removing the vapor from the evaporator and to raise its temperature and pressure to a point such that it (vapor) can be condensed with normally available condensing media is done by
 - A) throttle valve
 - B) condenser
 - C) compressor
 - D) evaporator
18. If the vapor is superheated after compression, the operation is called as
 - A) subcooled compression
 - B) wet compression
 - C) dry compression
 - D) superheated compression
19. In an unsaturated air the state of vapor is
 - A) wet
 - B) saturated
 - C) superheated
 - D) unsaturated
20. During sensible heating of moist air, enthalpy
 - A) increases
 - B) decreases
 - C) remains same
 - D) may increase or decrease

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

	Marks	BT*	CO*	PO*
1. a) Define the following the following with suitable examples: i) Open system ii) Closed system iii) Isolated system iv) Thermodynamic process	8	L*1	1	1
b) State the first law of thermodynamics applied to a system undergoing a process and a cycle, also prove that the total energy is a property of the system.	8	L2	2	1
2. a) Derive the expression for work transfer in i) Isothermal system ii) Isobaric system.	8	L2	1	1
b) Give thermodynamic definitions of specific heat and constant pressure (C_p) and specific heat at constant volume (C_v). Also show that the difference between them equals characteristic gas constant (R)	8	L2	2	1
3. a) State and explain Zeroth law of Thermodynamics.	4	L1	1	1
b) What do you mean by Thermodynamic equilibrium? Explain.	4	L1	1	1

- c) Give Kelvin Planck's and Clausius' statements of second law. Also prove their equivalence.

8 L2 2 1

Unit – II

4. a) Draw the P-V diagram for a pure substance whose volume increases during melting. Indicate all the salient points/regimes.
 b) In an air standard Diesel cycle, the compression ratio is 16. At the beginning of isentropic compression, the temperature is 15°C and pressure is 0.1 MPa. Heat is added until the temperature at the end of constant pressure process is 1500°C. Calculate
 i.) cut-off ratio
 ii.) heat supplied per kg of air
 iii.) cycle efficiency

8 L1 3 1

8 L2 4 1

5. a) Steam expands isentropically from 60 bar and 500°C to 100 kPa in a steam turbine. Calculate the dryness fraction and enthalpy of steam after expansion. If the steam flows steadily through the turbine with a mass flow rate of 100 kg/min, determine the turbine work. Neglect heat losses, potential, and kinetic energy effects.
 b) Explain the effect of following on performance of Rankine cycle:
 i) Lowering condenser pressure
 ii) Superheating steam to higher temperature

8 L3 3 2

8 L1 4 1

6. a) Explain the processes in an Ideal Rankine cycle with schematic and T-S diagrams.
 b) Derive the expression for air standard efficiency of Otto cycle.

8 L1 3 1

8 L2 4 1

Unit – III

7. a) A vapour compression refrigeration system of 10 ton capacity uses R-12 and operates between condenser and evaporator temperatures of 40°C and –10°C respectively, determine:
 i) Refrigerant flow rate
 ii) COP
 iii) Power required to drive the compressor.
 Take C_p of superheated refrigerant = 0.7467 kJ/kgK.

10 L3 5 2

- b) List the desirable properties of Ideal refrigerant.

6 L1 5 1

8. a) Sketch and explain the working of
 i.) Summer air conditioning system
 ii.) Winter air conditioning system
 b) Consider 100m³ of air water vapour mixture at 0.1 MPa, 35°C, 70% relative humidity. Calculate
 i) Humidity ratio
 ii) Dew point temperature
 iii) Mass of air
 iv) Mass of water vapour

10 L1 5 1

6 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
